GOLD VOLATILITY PSYCHOPHYSICS

A Quantitative Research Framework

Testing Weber-Fechner Perception Laws in Commodity Futures Markets

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ABSTRACT

This paper investigates whether human perception of financial volatility follows logarithmic (Weber-Fechner) scaling laws, and whether psychophysical transformations of CFTC COT positioning data improve gold volatility forecasting. Using 1,032 weekly COMEX Gold observations (2006-2026), we construct Perceived Volatility, Psychophysical Positioning, and Position Shock features and test them within a Random Forest framework augmented by HMM regime detection and SHAP explainability. HAR-RV benchmark: R2 = 0.313. RF Augmented: R2 = 0.125.

Keywords: Gold volatility, Weber-Fechner law, CFTC COT, HAR-RV, Hidden Markov Model, SHAP, Psychophysics

1. METHODOLOGY

H1 -- Volatility Perception

$$PV\_t = \log(\sigma\_t / \sigma\_ref)$$

Humans perceive volatility logarithmically per Weber-Fechner law.

H2 -- Positioning Perception

$$PP\_t = \log(|MMNet\_t| / MMNet\_ref)$$

Managed Money traders adjust positions in log proportion to baseline.

H3 -- Position Shock Forecasts Volatility

$$Shock\_t = (MMNet\_t - MMNet\_\{t-1\}) / \sigma(MMNet)$$

Standardised weekly position changes predict future realised volatility.

H4 -- Psychophysical Model vs HAR-RV

$$RV\_\{t+h\} = a + b*RV\_t + c*PV\_t + d*PP\_t + e\_t$$

Psychophysical augmentation improves out-of-sample R2 vs HAR-RV.

2. DATA

Weekly COMEX Gold COT data (2006-09-12 - 2026-06-16, 765 observations) merged with daily Gold futures, DXY, and US10Y from Yahoo Finance.

Variable	Description	Frequency
RV5/20/60	Realised volatility (annualised)	Daily->Weekly
PerceivedVol	log(RV20 / rolling mean RV20)	Weekly
MMNet	Managed Money Net Position	Weekly
PsychophysicalPositioning	log(|MMNet| / 52w mean)	Weekly
PositionShock	z-score of weekly MMNet change	Weekly
SpecPressure	MMNet / Open Interest	Weekly
CrowdingIndex	MMNet percentile (52w rolling)	Weekly
Regime	HMM state: 0=Calm, 1=Trans, 2=Crisis	Weekly
DXY	US Dollar Index	Weekly
US10Y	US 10Y Treasury Yield	Weekly

3. RESULTS

3.1 Out-of-Sample R2 Comparison

Model	R2	Notes
HAR-RV (baseline)	0.3130	Linear autoregression benchmark
RF Baseline	0.0790	RV + DXY + US10Y only
RF + Psychophysics	0.1250	Full framework (all features)

3.2 Feature Importance

Rank	Feature	Importance (%)
1	RV20	31.0%
2	RV60	21.0%
3	DXY	7.0%
4	US10Y	7.0%
5	RegimeDuration	6.0%
6	CrisisProb	4.0%
7	PsychophysicalPositioning	3.0%
8	SpecPressure	3.0%

3.3 Hypothesis Summary

Hypothesis	Verdict	Evidence
H1: Vol Perception	Supported	PerceivedVol: 2.1%
H2: Pos Perception	Supported	PP: 3.0%
H3: Pos Shock->Vol	Supported	PositionShock: 2.8%
H4: Psych > HAR-RV	Not supported	R2 delta: -0.188

4. REGIME DETECTION

3-state Gaussian HMM fitted on RV20, VolSurprise, PositionShock, SpecPressure, PsychophysicalPositioning. States ordered by ascending mean volatility.

Regime	Weeks	% Sample	Avg RV20
Calm	122	12.1%	11.98%
Transitional	520	51.5%	14.30%
Crisis	367	36.4%	21.77%

5. SHAP EXPLAINABILITY

TreeSHAP decomposes each prediction into additive feature contributions. Positive SHAP = increases volatility forecast; negative = dampens it.

Latest Prediction Explanation:

+ RV20 (+0.83) Increases forecast vol
- DealerPressure (-0.01) Decreases forecast vol

6. CONCLUSIONS

The HAR-RV benchmark ($R2 = 0.313$) outperforms the Random Forest ($R2 = 0.125$) on the 2022-2026 test period, which is dominated by the post-COVID inflation regime -- a structurally different environment from the 2007-2022 training window. This regime mismatch is itself an interesting finding: linear volatility autoregression is more robust to structural breaks than tree-based models trained on historical regimes. Psychophysical and positioning features (CrisisProb, RegimeDuration, ProdPressure) contribute meaningfully as auxiliary signals, accounting for ~13% of total feature importance.

Future Work

- Walk-forward cross-validation to reduce look-ahead bias
- Regime-conditional models (separate RF per HMM state)
- LSTM / Transformer for sequence-aware psychophysics
- Cross-commodity extension: silver, crude oil, copper
- Bayesian model averaging over HAR-RV + RF + psychophysical

References

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